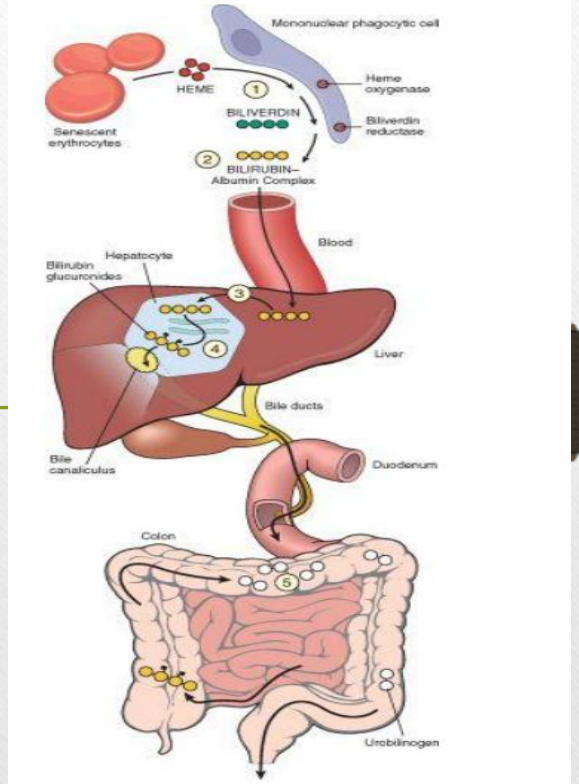
METABOLISM OF BILIRUBIN and PORPHYRIA

Dr. Humaira Aman Ali

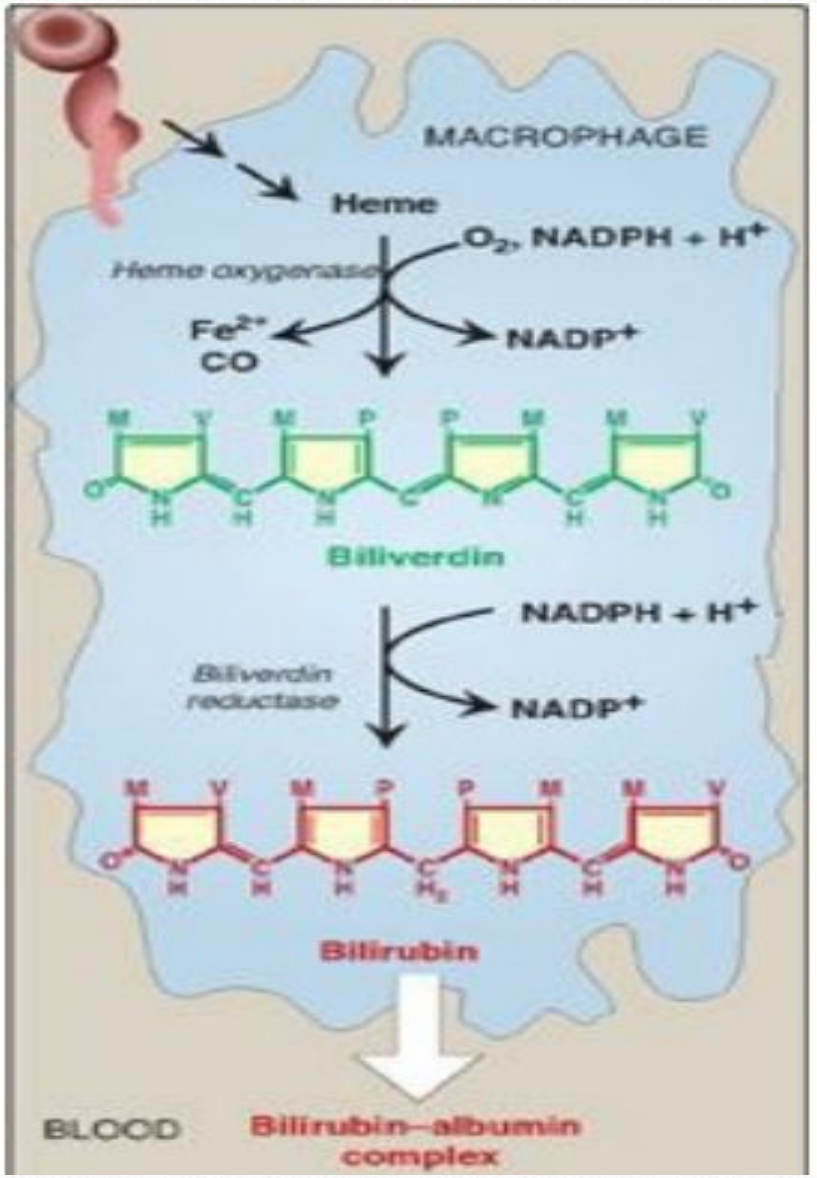


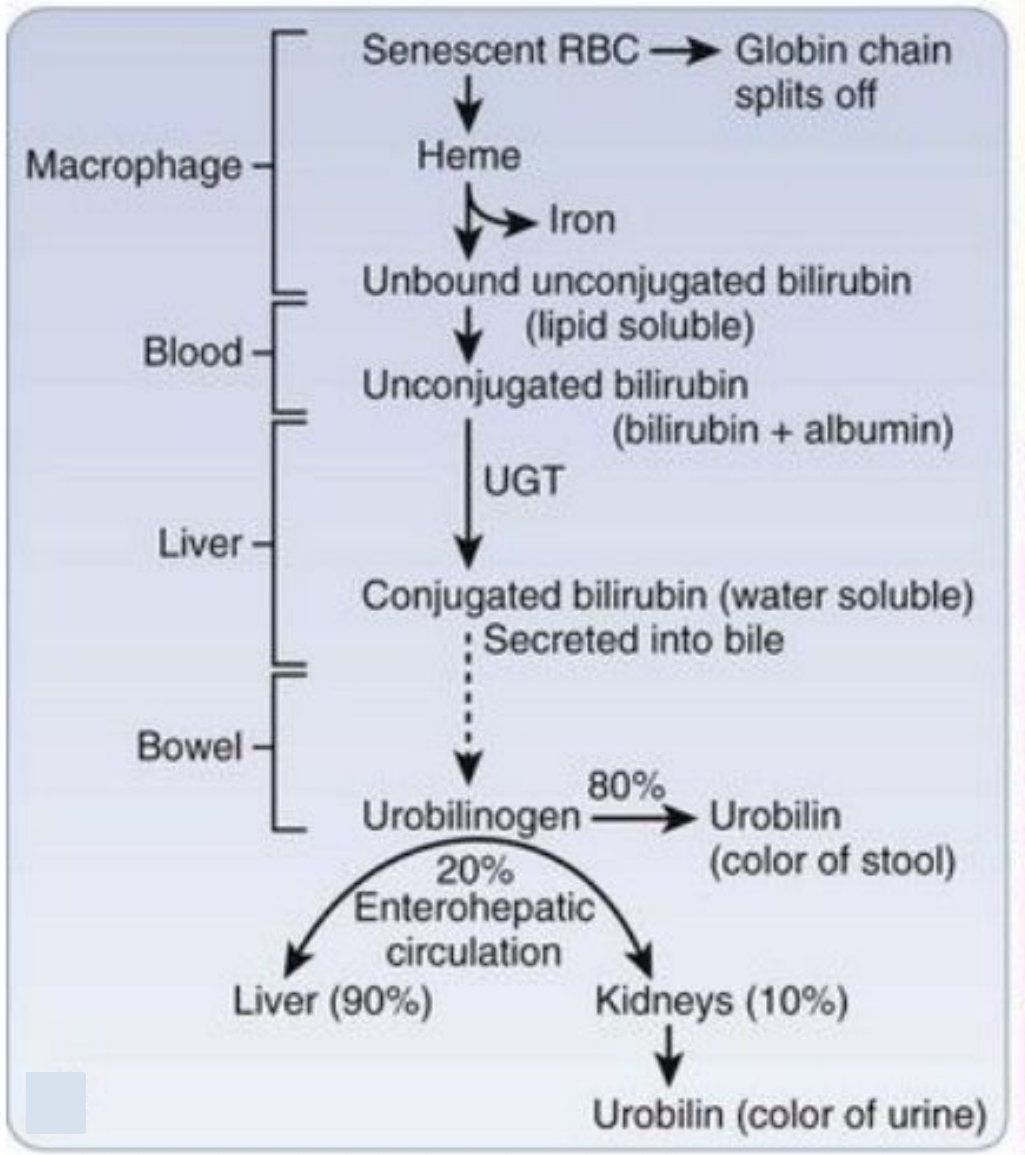
METABOLISM OF BILIRUBIN

Degradation of Heme:

1. Formation of bilirubin:

- Bilirubin is the end product of heme degradation.
- The majority of daily production (85%) is derived from breakdown of senescent red cells by the mononuclear phagocytic system, especially in the spleen, liver, and bone marrow.
- Most of the remainder (15%) of bilirubin is derived from the turnover of hemoproteins and from premature destruction of red cell precursors in the bone marrow.
- Whatever the source, intracellular **heme oxygenase** oxidizes **heme** to **biliverdin**, which is immediately reduced to **bilirubin** by **biliverdin reductase**.
- Bilirubin thus formed outside the liver is released and bound to serum albumin.


MACROPHAGE
Heme
O_2, NADPH + H^+
Heme oxygenase
Fe^{2+}
CO
$NADP^+$
Biliverdin
NADPH + H^+
Biliverdin reductase
$NADP^+$
Bilirubin
BLOOD
Bilirubin–albumin complex

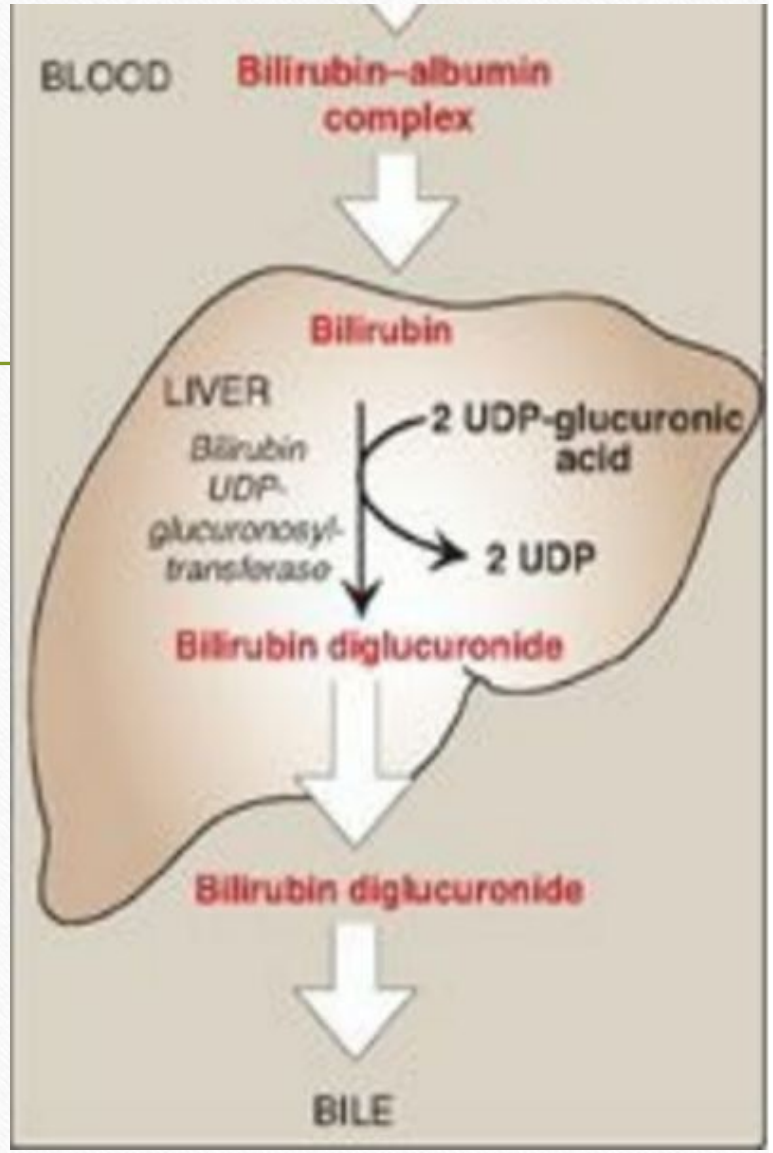

Senescent RBC
Globin chain splits off
Heme
Iron
Macrophage
Unbound unconjugated bilirubin (lipid soluble)
Blood
Unconjugated bilirubin (bilirubin + albumin)
UGT
Liver
Conjugated bilirubin (water soluble)
Secreted into bile
Bowel
Urobilinogen
80%
Urobilin (color of stool)
20%
Enterohepatic circulation
Liver (90%)
Kidneys (10%)
Urobilin (color of urine)

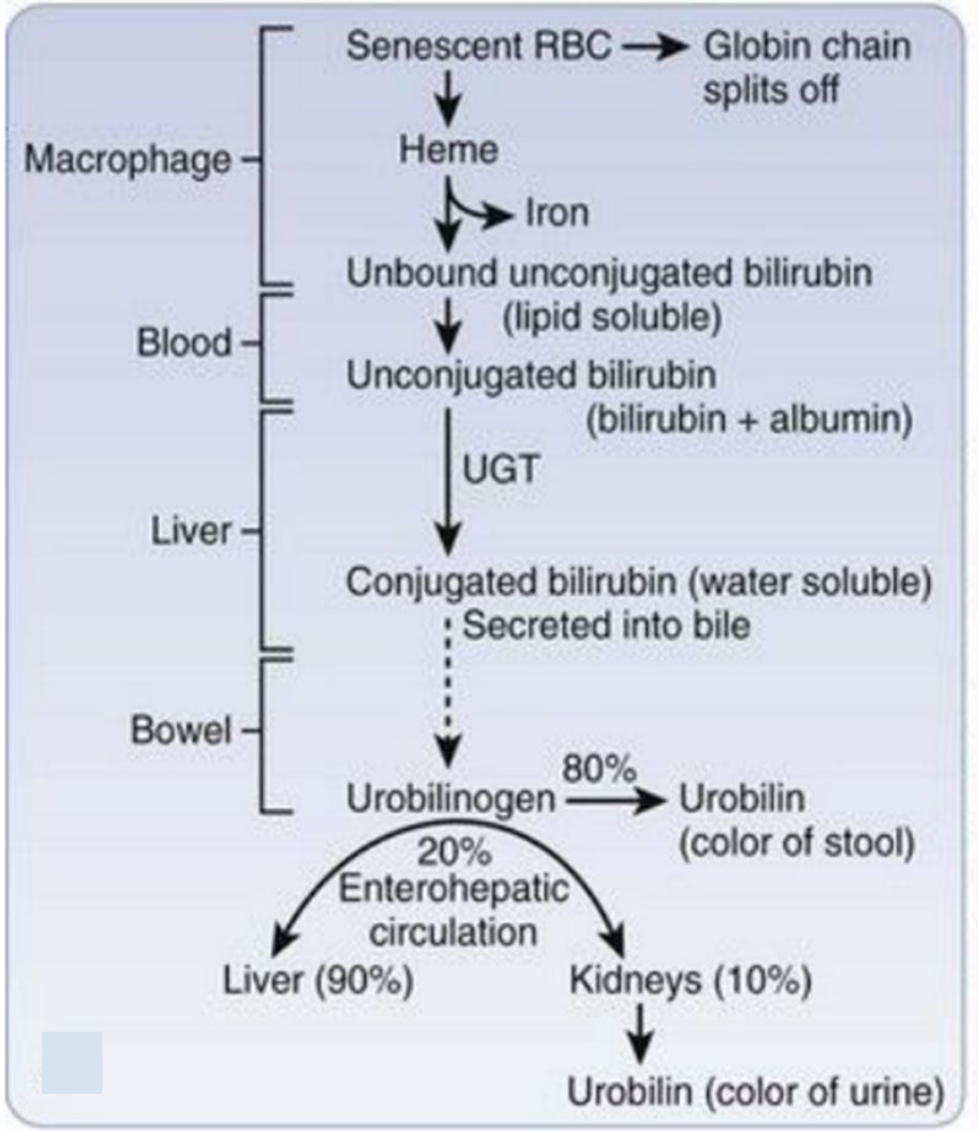
2. Uptake of bilirubin by the liver:

- Hepatic processing of bilirubin involves carrier mediated uptake at the sinusoidal membrane.
- Certain anionic drugs, such as salicylates and sulfonamides, can displace bilirubin from albumin, permitting bilirubin to enter the central nervous system (CNS). This causes the potential for neural damage in infants.

3. Formation of bilirubin diglucuronide:

- ***Conjugation:*** In the hepatocyte, the solubility of bilirubin is increased by the addition of two molecules of glucuronic acid, producing **bilirubin diglucuronide**. This process is referred to as conjugation.
- The reaction is catalyzed by microsomal **bilirubin uridine diphosphate (UDP)-glucuronosyltransferase (bilirubin UGT or UGT1A1).**
- Varying degrees of deficiency of bilirubin UGT result in **Crigler-Najjar I and II and Gilbert syndrome.**


BLOOD
Bilirubin–albumin complex
Bilirubin
LIVER
Bilirubin UDP-glucuronosyl-transferase
2 UDP-glucuronic acid
2 UDP
Bilirubin diglucuronide
Bilirubin diglucuronide
BILE

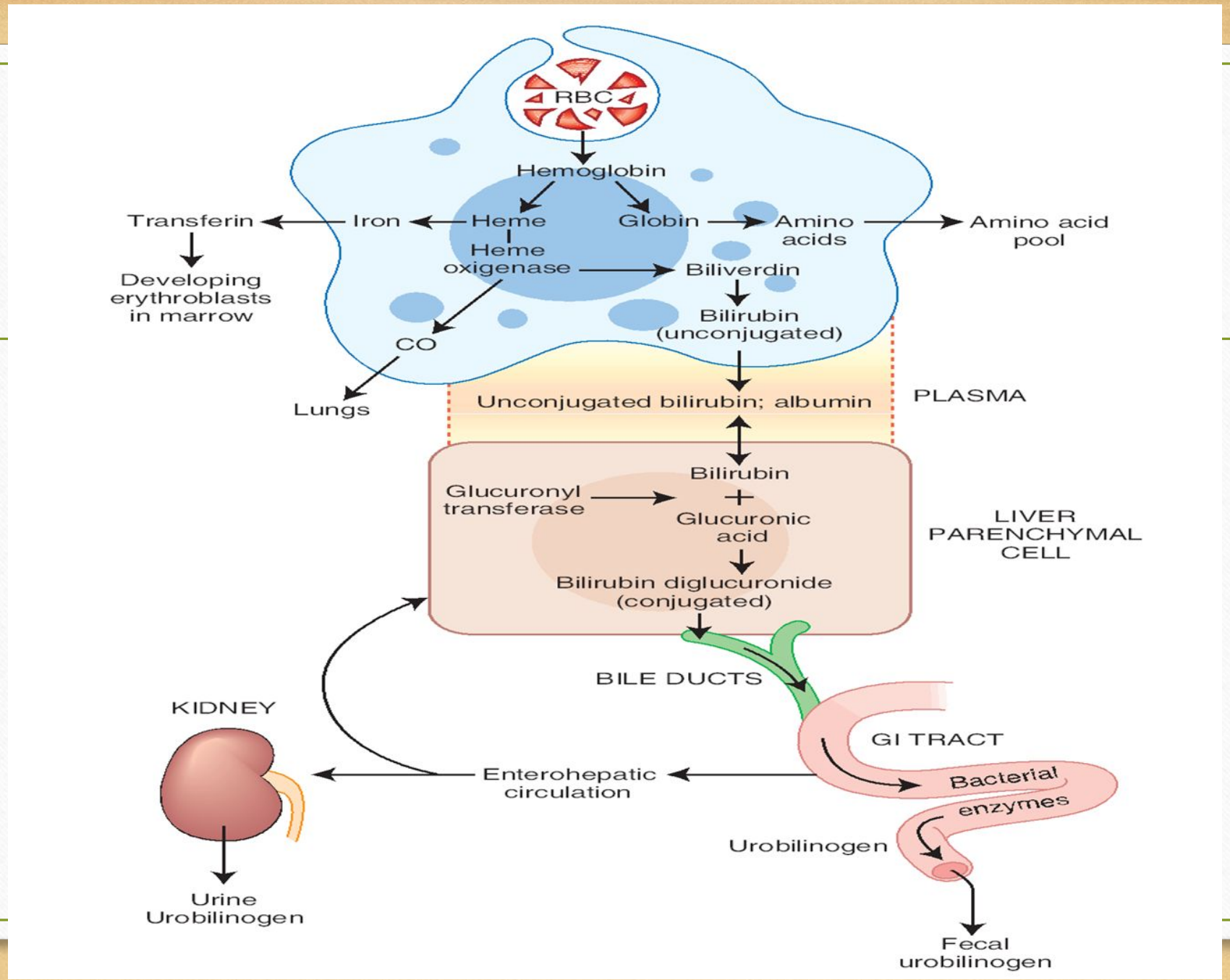

Macrophage
Senescent RBC
Globin chain splits off
Heme
Iron
Blood
Unbound unconjugated bilirubin (lipid soluble)
Unconjugated bilirubin (bilirubin + albumin)
Liver
UGT
Conjugated bilirubin (water soluble)
Secreted into bile
Bowel
Urobilinogen
80%
Urobilin (color of stool)
20%
Enterohepatic circulation
Liver (90%)
Kidneys (10%)
Urobilin (color of urine)

4. Secretion of bilirubin into bile:

- Water-soluble Bilirubin diglucuronide (conjugated bilirubin-CB) is actively excreted into the bile.
- This step is susceptible to impairment in liver disease.
- A rare deficiency in the protein required for transport of CB out of the liver results in Dubin-Johnson syndrome.

5. Formation of urobilins in the intestine:

- Most bilirubin glucuronides are deconjugated in the gut lumen by bacterial β-glucuronidases and to yield **UROBILINOGEN,** a colorless compound.
- Most of the urobilinogen is oxidized by intestinal bacteria to stercobilin, which gives feces the characteristic brown color.
- Approximately 20% of the urobilinogens formed are reabsorbed in the ileum and colon and enters the portal blood, returned to the liver, and re-excreted into bile. Known as enterohepatic urobilinogen cycle.
- The remainder of the urobilinogen is transported by the blood to the kidney, where it is converted to yellow urobilin and excreted, giving urine its characteristic color.


RBC
Hemoglobin
Transferin
Iron
Heme
Globin
Amino acids
Amino acid pool
Heme oxigenase
Biliverdin
Developing erythroblasts in marrow
Bilirubin (unconjugated)
CO
Lungs
Unconjugated bilirubin; albumin
PLASMA
Bilirubin
Glucuronyl transferase
Glucuronic acid
LIVER PARENCHYMAL CELL
Bilirubin diglucuronide (conjugated)
BILE DUCTS
KIDNEY
GI TRACT
Enterohepatic circulation
Bacterial enzymes
Urobilinogen
Urine Urobilinogen
Fecal urobilinogen

Hereditary Hyperbilirubinemias

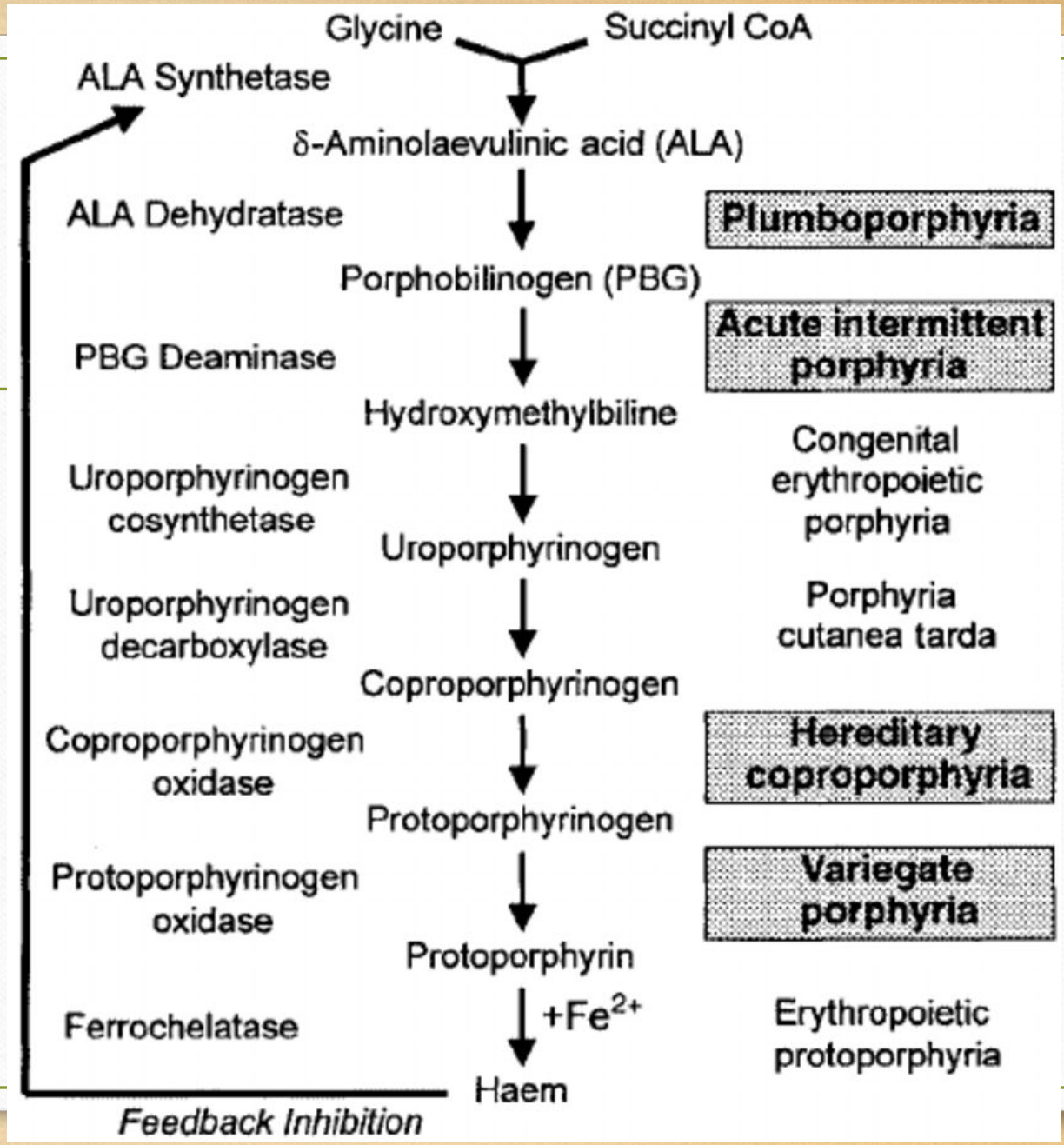
- Multiple genetic mutations can cause hereditary hyperbilirubinemia.
- In Crigler-Najjar syndrome type I hepatic UGT1A1 is completely absent.
- Serum unconjugated bilirubin reaches very high levels, producing severe jaundice and icterus.
- Without liver transplantation, this condition is invariably fatal, causing death secondary to kernicterus within 18 months of birth.

Disorder	Inheritance	Defects in Bilirubin Metabolism	Liver Pathology	Clinical Course
UNCONJUGATED HYPERBILIRUBINEMIA				
Crigler-Najjar syndrome type I	Autosomal recessive	Absent UGT1A1 activity	None	Fatal in neonatal period
Crigler-Najjar syndrome type II	Autosomal dominant with variable penetrance	Decreased UGT1A1 activity	None	Generally mild, occasional kernicterus
Gilbert syndrome	Autosomal recessive	Decreased UGT1A1 activity	None	Innocuous
CONJUGATED HYPERBILIRUBINEMIA				
Dubin-Johnson syndrome	Autosomal recessive	**A defect in hepatocellular excretion of bilirubin glucuronides across the canalicular membrane.**	Pigmented cytopasmic globules;	Innocuous
Rotor syndrome	Autosomal recessive	Decreased hepatic uptake and storage? Decreased biliary excretion?	None	Innocuous

Porphyrias

- Porphyrias refers to a group of inherited (or occasionally acquired) defects in heme synthesis, resulting in the accumulation and increased excretion of porphyrins or porphyrin precursors.
- “Porphyria” refers to the red-blue color caused by pigment-like porphyrins in the urine of some patients.
- Each porphyria results in the accumulation of a unique pattern of intermediates caused by the deficiency of an enzyme in the heme synthetic pathway.

Heme synthesis



CLASSIFICATION

- The porphyrias are classified depending on whether the enzyme deficiency occurs in the erythropoietic cells of the bone marrow or in the liver in:
  - **Erythropoietic porphyrias**
  - **Hepatic porphyrias:** Hepatic porphyrias can be further classified as
    - **Chronic hepatic porphyrias**
    - **Acute hepatic porphyrias**

Clinical Manifestation

- In general, individuals with an enzyme defect prior to the synthesis of the tetrapyrroles manifest **abdominal and neuropsychiatric signs,** whereas those with enzyme defects leading to the accumulation of tetrapyrrole intermediates show **photosensitivity** (that is, their skin itches and burns [pruritus] when exposed to visible light).

LEAD POISONING

- *Ferrochelatase* and *ALA dehydratase (ALAD)*[1] are particularly sensitive to inhibition by lead.
- Protoporphyrin and ALA accumulate in urine.
- *ALAD* deficiency porphyria is a very rare acute porphyria.

ERYTHROPOIETIC PROTOPORPHYRIA (EPP)

- This chronic disease is caused by a deficiency in *ferrochelatase*.
- Protoporphyrin accumulates in erythrocytes, bone marrow, and plasma.
- Patients are photosensitive.

ACUTE INTERMITTENT PORPHYRIA (AIP)

- This acute disease is caused by a deficiency in *hydroxymethylbilane synthase*[2].
- Porphobilinogen and ALA accumulate in the urine.
- Urine darkens on exposure to light and air.
- Patients are not photosensitive.

VARIEGATE PORPHYRIA (VP)

- This acute disease is caused by a deficiency in *protoporphyrinogen oxidase*.
- Protoporphyrinogen IX and other intermediates prior to the block accumulate in the urine.
- Patients are photosensitive.

HEREDITARY COPROPORPHYRIA (HCP)

- This acute disease is caused by a deficiency in *coproporphyrinogen III oxidase*.
- Coproporphyrinogen III and other intermediates prior to the block accumulate in the urine.
- Patients are photosensitive.

PORPHYRIA CUTANEA TARDA (PCT)

- This chronic disease is caused by a deficiency in *uroporphyrinogen decarboxylase*.
- Uroporphyrin accumulates in the urine.
- It is the most common porphyria.
- Patients are photosensitive.

CONGENITAL ERYTHROPOIETIC PORPHYRIA (CEP)

- This chronic disease is caused by a deficiency in *uroporphyrinogen III synthase*.
- Uroporphyrinogen I and coproporphyrinogen I accumulate in the urine.
- Patients are photosensitive.

Heme

Fe^{2+}

Protoporphyrin IX

Protoporphyrinogen IX

Coproporphyrinogen III

MITOCHONDRIA

CYTOSOL

Succinyl coenzyme A + Glycine

δ-Aminolevulinic acid

δ-Aminolevulinic acid (ALA)

Porphobilinogen

Hydroxymethylbilane

Uroporphyrinogen III —Spontaneous→ Uroporphyrin III

Coproporphyrinogen III —Spontaneous→ Coproporphyrin III

Uroporphyrinogen I —Spontaneous→ Uroporphyrin I

Coproporphyrinogen I —Spontaneous→ Coproporphyrin I

KEY:

Hepatic porphyria

Erythropoietic porphyria





THANK YOU